



Read Deeper, *Not Faster.*

A learning-analytics system for detecting and interrupting mindless reading in digital environments.

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41

0%
of digital learning time
is spent off-task

63% of these episodes are **aware**
— learners know they've drifted,
but cannot redirect.

- Bühler et al.

THE PROBLEM, STATED

Remote learners lose more than a third of their study time to a cognitive failure they often cannot perceive. Bühler et al. (Temporal Dynamics of Meta-Awareness of Mind Wandering During Lecture Viewing: Implications for Learning and Automated Assessment Using Machine Learning) show that in digital learning, learners' minds are task-unrelated for roughly 41% of probed moments, often preoccupied with hunger, fatigue, or personal worries. Worse, 63% of these episodes are **aware**: the learner notices they have drifted, but cannot or does not redirect. The remaining shares are **unaware zone-outs**, where the mind has disengaged entirely while the eyes continue moving across text.

Montana App targets this detection gap specifically in asynchronous, text-based digital reading, where no instructor can notice, no peer can interrupt, and the learner has no reliable feedback signal about when their own attention has collapsed.

2 Population Who Experiences It

WHO IS AFFECTED

Montana App is designed for **remote and hybrid learners engaged in self-paced, text-heavy digital reading**: undergrad/ graduate students, online course participants, and independent professionals working through articles, chapters, and PDFs without a live instructor. These learners face structural disadvantages: no external pacing, no classroom accountability, no social norm of visible attentiveness, and critically, *no reliable feedback about when their own comprehension has collapsed*.

VULNERABLE GROUPS

Online course students

Upskilling professionals

Graduate self-study

Homeschool/distance learners

THE ZONE-OUT CLUSTER

Our primary target

Across 15 probes in a single lecture session, zone-out learners report on average:



The steepest learning penalty. Bühler et al. found zone-out cluster members show significantly lower performance on **both fact-based memory and inference learning**, a more severe deficit than any other mind-wandering subtype.

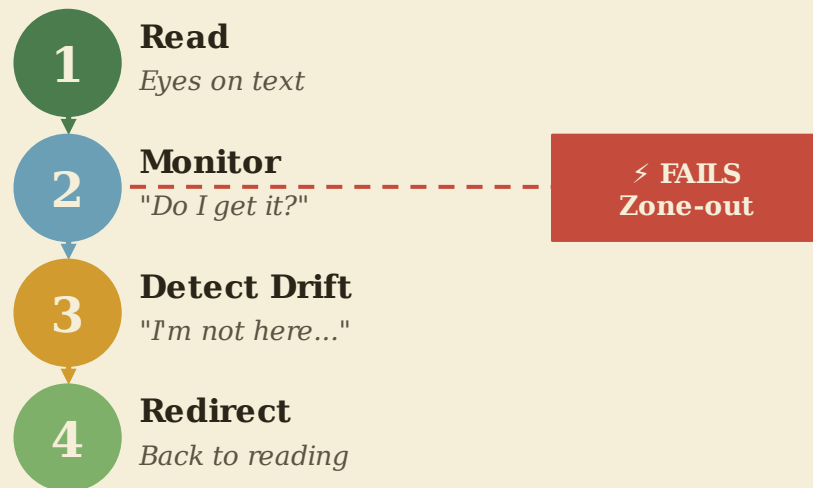
Least likely to self-recover without scaffolding

3 The Learning Process Involved

At stake is **metacognitive monitoring during self-regulated reading**; the learner's internal capacity to ask "Am I still following this?" and redirect when the answer is no. When this monitoring loop fails, the surface behavior of reading continues, eyes moving, scroll advancing, while no meaningful encoding occurs. Critically, Bühler et al. found that excessive monitoring itself **harms** deep understanding.

Montana, therefore, aims not to maximize self-monitoring, but to deliver formative questions or a calibrated nudge— a short story at the right moment.

THE MONITORING LOOP



⚠ **The Paradox:** excessive Step 2 monitoring itself reduces deep understanding (Bühler et al.). A calibrated nudge, not constant interruption, is the goal.

4 Environment Where It Occurs



Read Deeper, Not Faster

Montana monitors your reading pace in real time. When you scroll past comprehension, the forest intervenes — gently, wisely, like a patient guide.

⌵ Scroll velocity tracking

📖 AI comprehension quizzes

🌲 Contextual intervention stories

Start Reading

Reading title

e.g. The Pragmatic Programmer, Chapter 3

📄 Paste Text

📄 Upload PDF

Paste your article, essay, chapter, or any long-form text here...

📖 Begin Mindful Reading

SCREENSHOT · montana-app

MONTANA APP | READING INTERFACE

A responsive web and mobile application (Next.js, Supabase). Learners upload a PDF or paste text; the app chunks content into 10–15 minute sessions, the window in which task-unrelated thoughts empirically surge (Bühler et al.). The *forest aesthetic* is not decorative: Berman, Jonides, and Kaplan (2008) demonstrated that even viewing natural imagery improves directed-attention abilities by roughly 20% on the backwards digit-span task.

INTERFACE ELEMENTS

- 1 Parchment reader (cream #F8F3E9)**
Reduces eye strain vs white screen
- 2 Ambience toggle (forest audio)**
ART-grounded attention restoration
- 3 Paste text / Upload PDF**
Content processed into 10–15 min chunks
- 4 Begin Mindful Reading**
Starts session, initiates scroll tracking

5a Theory: Dual-Type Mind Wandering



Montana's detection model is grounded in the research program of **Bühler, D'Mello, and colleagues** on meta-awareness of mind wandering in digital learning (Bühler et al., 2024). This research makes a clinically important distinction between two forms of task-unrelated thought that older research often conflated, forms that have ***distinct neural signatures, distinct learning costs, and require distinct interventions.***

ZONE-OUT

unaware MW

No meta-awareness. The learner does not know their mind has wandered. Eye movements become fast and short; blinks lengthen; the brain has disengaged while the eyes continue scanning.

Learning damage (vs. on-task):

- ✗ **Fact-based memory**
- ✗ **Inference / deep understanding**

→ **Intervention: BREAK /nature-walk prompt (Berman 2008)**

TUNE-OUT

aware MW

Has meta-awareness. The learner knows they've drifted but cannot or will not redirect. Eye movements slow; gaze drifts off-screen; the attention has decoupled from the text.

Learning damage (vs. on-task):

- ✓ **Fact memory (mostly preserved)**
- ✗ **Inference / deep understanding**

→ **Intervention: short story reset/coffee break**

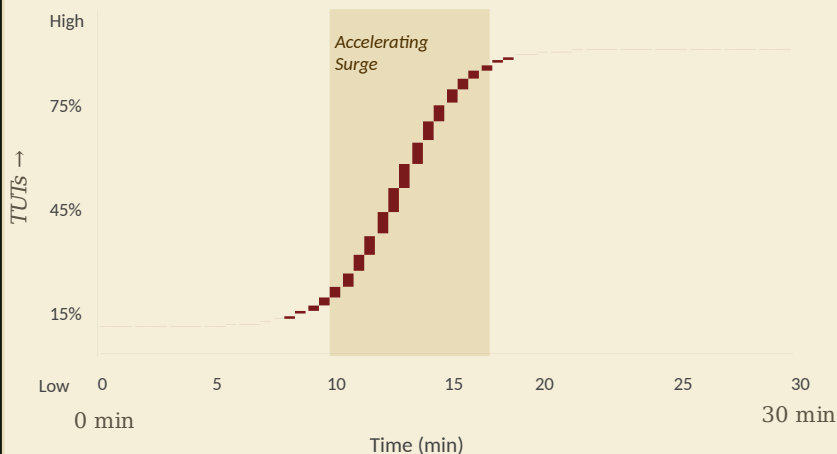
5b

Theory: Temporal Dynamics & Detection



TEMPORAL DECAY

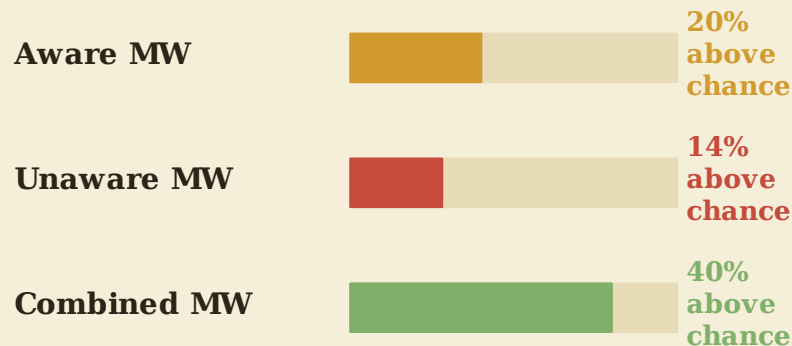
The 10-15 minute window



Task-unrelated thoughts surge sharply between 10 and 15 minutes into a session (*Bühler et al.*). **Montana chunks all content at this empirical boundary.**

DETECTION EVIDENCE

Multimodal > Unimodal



Fusing eye-tracking + facial video + physiology, Bühler et al. (2024) achieved **40% above-chance detection on combined MW**. Critically, webcam-based gaze (Zhao et al., 2017) performs near-equal to dedicated eye trackers, making scalable, affordable deployment feasible.

6 AIM: What Montana App Does



Montana App captures behavioral signals of mindless reading by scroll velocity, quiz accuracy, and (optionally) webcam-based gaze and delivers evidence-based scaffolding *before the session is wasted*.

FREE TIER · scroll + quiz

- Real-time scroll velocity (px/s)
- Auto-generated formative quiz after each chunk
- Dual-metric zone-out flag: SVR + QAS
- Intervention: short story OR walk prompt

PAID TIER · + webcam gaze

- All free-tier features, plus:
- Webcam gaze: saccade, blink, amplitude
- GZP: gaze zone-out probability score
- Mid-chunk detection (no quiz wait)

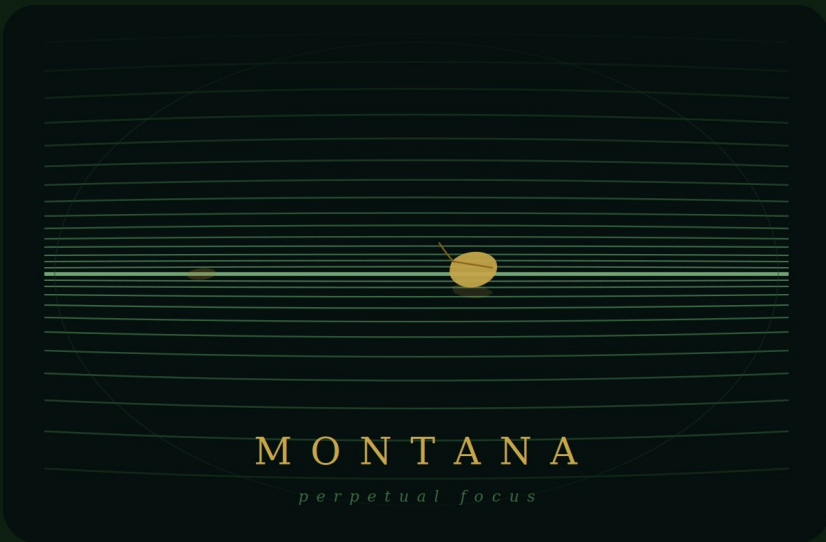
DESIGN PRINCIPLES

1. Non-intrusive Fires only when both signals cross threshold

2. Learner agency Offers choice; every dismiss is logged

3. Scientifically grounded Bühler (2024) + Berman (2008) & other back every intervention

7 Opportunity: The Product



A complete product, captured here as it exists today.

Montana App is a responsive Next.js web application deployed publicly (ongoing). The learner uploads a PDF or pastes raw text; the LLM back-end chunks text into 10-15 minute reading sessions of ~2,000-2,500 words each. A forest soundscape plays optionally; a parchment reader opens on the next screen.

As the learner reads, JavaScript samples

scroll velocity every 500ms in pixels per second. When the chunk ends, the app generates 3-5 comprehension questions with Gemini/OpenAI and presents them as a non-dismissible overlay. Scroll velocity and quiz accuracy together form the behavioral substrate.

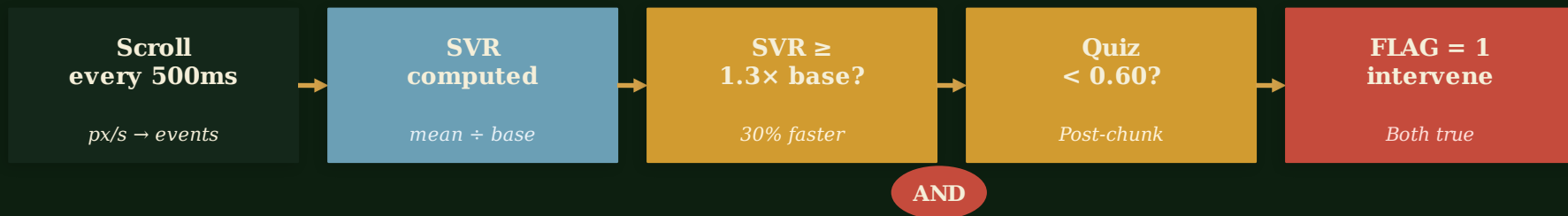
A three-day initial and later cumulative calibration phase establishes each learner's baseline scroll velocity and quiz accuracy under focused reading, transforming raw metrics into personalized deviation scores. When deviations exceed the threshold, Montana intervenes with an AI-generated short story (related to the text) or a five-minute walk prompt.

For paid subscribers, the device webcam adds a third channel: real-time gaze features feed a zone-out probability score, enabling mid-chunk detection without waiting for the quiz. Location and ambient noise are logged alongside each session to build a personalized focus-environment map, positioned as a correlational *roadmap feature*.

8 Logic: IF Evidence = 1, THEN Act

The detection logic runs in two tiers depending on subscription. For free-tier learners, the flag requires corroboration from two independent behavioral channels; for paid-tier, a third gaze-based channel can fire independently. This deliberate design minimizes the false positives that would otherwise erode learner trust in the system over time (Bühler et al., 2024).

FREE TIER · Dual-Metric Zone-out Flag



PAID TIER · Gaze-Augmented Zone-out Probability (GZP)



Severity escalation within session: Flag 1 → quiz/gentle overlay · Flag 2 → 2-3 min AI short story (topic-tied) · Flag 3+ → 5-min walk prompt (Berman et al., 2008)



Montana's dynamic logic engine distinguishes conceptual struggles from attentional failures, providing a calibrated nudge instead of a constant interruption.

OBSERVED PATTERN (IF)	MONTANA ACTION (THEN)	PEDAGOGICAL LOGIC (WHY)
Potential Focus Decay (after 10–15 min)	Ask Formative Question on the text	Realigns Focus: 10–15 min is when decay surges. Repeating content improves retention even with imperfect prediction. (Hutt et al., 2021)
Focus Maintained (SVR/GZP within baseline)	Delay Formative Question: Extend session beyond 10–15 min	Respects Flow: Formative questions act as a cognitive break only when decay is detected.
High Accuracy & High Focus	Positive Affirmation: "Mastery confirmed! You've covered this section perfectly."	Combats Paradox: Students suffer meta-awareness 16% of the time. Affirming understanding stops the learner from worrying, which harms deep comprehension.
Low Accuracy & High Focus	Content Clarification: Engaging explanation, summary, or targeted hint	Conceptual Scaffolding: Helps create a mental model of the text to specifically avoid the "tune-out" condition.
Low Accuracy & Low Focus	Narrative Reset: AI Short Story or Coffee/Walk Break	Mindless Reading: Addresses Unaware Mind Wandering (Zone-out); total reset restores executive control.
Long-term Contextual Data (3-day baseline)	Focus Zone Mapping: Environmental advice for time and location	Environment Optimization: Aligns task difficulty with the learner's biological peak attention and ideal ambient conditions.

8b Building Intrinsic Mastery: Focus Muscle Challenges

To prevent system-dependence, Montana includes gamified challenges to help learners develop their own focus habits.

GAMIFIED FOCUS CHALLENGES

1

Deep Focus Sprint

Earn points for maintaining a high focus score for half an hour.

2

No Phone Zone

Rewards the ability to read for 45 minutes without checking external devices.

3

Marathon Reader

A challenge for completing three dense chapters in a single session.

Focus Challenges frame Montana interventions as temporary scaffolding, building the learner's capacity over time.



Montana serves as a laboratory for Design-Based Empirical Research, investigating the efficacy of narrative transport as a cognitive intervention.

THE THEORY

Narrative Transport

"Getting engaged in a short story" may act as a superior palate cleanser for the brain, resetting the executive control system more effectively than a simple prompt to "try harder."

THE RESEARCH GOAL

Post-Intervention Accuracy

Montana tracks scores after interventions. If student scores improve significantly more after a short story than after a text break, narrative re-engagement is a proven mechanism for focus restoration.

THE DATA STRATEGY

Prescriptive Story Matching

The system logs which story genres and lengths lead to the highest focus recovery, allowing the app to eventually prescribe targeted stories for specific learner profiles.

Hypothesis: Narrative re-engagement is a mechanism for focus restoration, trackable and prescribable at scale.

9

L1 - Raw Data: What Montana Logs



Every learner action generates a timestamped event. Data sits in Supabase (PostgreSQL) across four relational tables; *users* stores the learner profile and calibration baseline, *readings* holds uploaded content and chunk boundaries, *sessions* records one row per chunk attempt, and *events* is the analytics stream, every scroll sample, quiz response, gaze frame, and intervention outcome is appended with millisecond precision.

SAMPLE · events table, one learner, one chunk

event_type	timestamp	scroll_vel (px/s)	SVR	quiz_score	gaze_ZOP	blink_ms	FLAG
scroll_sample	14:03:12.500	148	1.02	—	—	—	0
scroll_sample	14:03:13.000	142	0.99	—	—	—	0
scroll_sample	14:09:44.000	389	2.70	—	—	—	0
gaze_sample	14:09:44.000	—	—	—	0.71	412	0
scroll_sample	14:12:10.000	372	2.58	—	—	—	0
quiz_response	14:18:04.000	—	—	0.40	—	—	1
intervention_shown	14:18:05.000	—	—	—	—	—	—

↑ Scroll spikes $2.7 \times$ baseline + gaze ZOP = 0.71 + quiz = 0.40 → FLAG fires → intervention triggered

KEY FIELDS LOGGED

scroll_velocity_px_s · quiz_score · gaze_zone_out_prob · blink_duration_ms · saccade_velocity_deg_s · saccade_amplitude_deg · location_tag, decibel_level

10 L2 - Feature Engineering



Each raw channel is transformed into a theorized cognitive indicator. The three detection features below map directly to documented behavioral signatures of mind wandering.

SVR <i>Scroll Velocity Ratio</i>	RAW <i>scroll_velocity_px_s ÷ baseline_scroll_velocity</i>	COMPUTE <i>Rolling mean(current chunk) ÷ personalized baseline</i>	TRIGGER $\geq 1.3\times$	MAPS TO SVR > 1.3 = scrolling 30%+ faster than own focused pace. The AND requirement with QAS prevents false positives from easy content.
QAS <i>Quiz Accuracy Score</i>	RAW <i>quiz_score (3-5 auto-gen Q's/chunk)</i>	COMPUTE <i>correct ÷ total per 10-15 min chunk</i>	TRIGGER < 0.60	MAPS TO QAS < 0.60 = measurable comprehension failure. LLM-generated, multiple-choice, fresh per session. Non-dismissible before advancing.
GZP <i>Gaze Zone-out Prob · Paid</i>	RAW <i>saccade_velocity_deg_s blink_duration_ms saccade_amplitude_deg</i>	COMPUTE <i>Logistic regression over 30-sec rolling window</i>	TRIGGER ≥ 0.65	MAPS TO Maps to zone-out gaze signature: fast movement, short distance, long blinks (Bühler et al., 2024). Fres independent of scroll/quiz.

BASELINE CALIBRATION: 3-day phase: *baseline_scroll_velocity* = mean(chunks where quiz ≥ 0.70). Ensures baseline reflects focused reading, not any first session.

11 L3 - Data Processing



SVR Processing

Scroll events fire every 500ms via JavaScript `window.scrollY` differentials. Each event logs `scroll_velocity_px_s` to the events table with timestamp and `session_id`.

At chunk completion, the backend computes `mean(scroll_velocity_px_s)` across the chunk and divides by the user's `baseline_scroll_velocity`. SVR is dimensionless.

Threshold: $SVR \geq 1.3$. Chosen because it represents 30% above a learner's own focused-reading pace, well outside normal variance (0.9–1.2 across calibration data).

QAS Processing

At each chunk boundary, the LLM (Gemini/OpenAI) generates 3–5 multiple-choice questions targeting both factual recall and inferential comprehension from the chunk's content.

The quiz is non-dismissible, the learner must respond before proceeding. QAS = proportion correct; questions regenerate per session to prevent memorization.

Fusion with SVR: Flag fires IF $SVR \geq 1.3$ AND $QAS < 0.60$. The AND is deliberate corroboration from two independent channels aligns with Bühler et al.'s (2024) finding that multimodal detection outperforms any single signal.

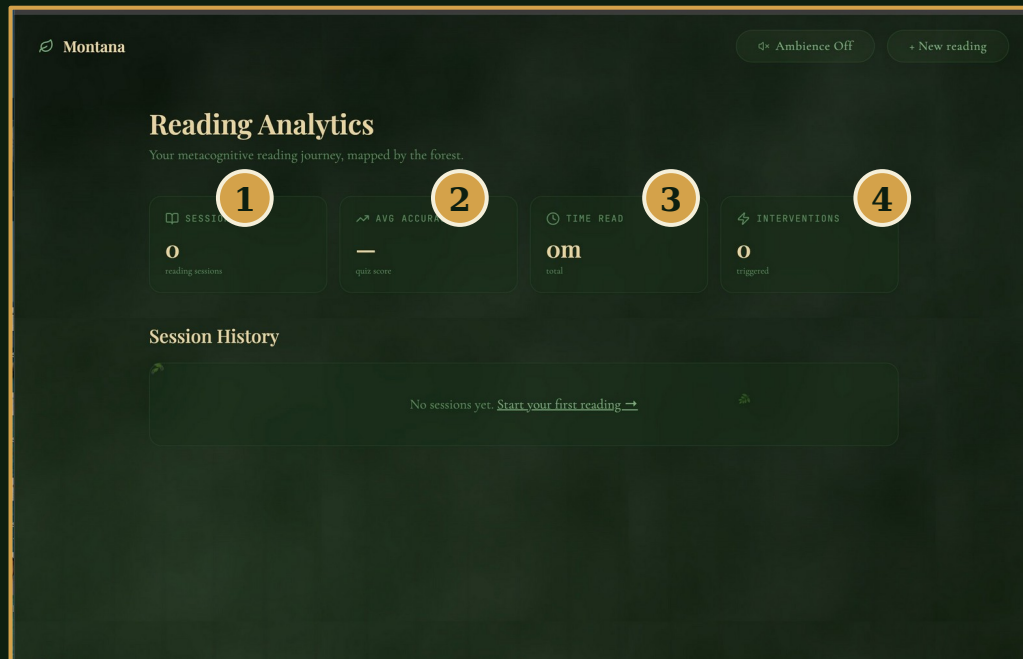
GZP Processing

Webcam frames are processed client-side using MediaPipe face mesh, no video is transmitted or stored. For every 30-second rolling window, four features are computed:

- *mean saccade velocity ($^{\circ}/s$)*
- *75th percentile blink duration (ms)*
- *75th percentile saccade amplitude ($^{\circ}$)*
- *vergence angle standard deviation*

These feed a logistic regression pre-trained on the Bühler et al. (2024) public dataset. GZP is the posterior probability of zone-out. **Threshold: $GZP \geq 0.65$.** Unlike the SVR+QAS flag, GZP can fire mid-chunk without waiting for the quiz boundary, enabling faster intervention for paid-tier users.

12 L4 - Focus Dashboard



CURRENT · empty state

WHAT THE LEARNER WILL SEE

After each session, the Reading Analytics dashboard surfaces four summary tiles (numbered at left) and below them a growing **Session History** with per-session detail: scroll profile, comprehension per chunk, intervention choices, and the personalized environment map.

ONCE POPULATED

- 1

14 sessions
- 2

72% avg
- 3

8h 43m
- 4

23 triggered

HOW THE LEARNER SHOULD READ THIS: *Low accuracy + high interventions + low session count signals early-stage struggle, Montana recommends environment changes before content changes. A rising accuracy trend is the primary indicator that scaffolding is working and can be gradually withdrawn.*

13 Cautions & Promise (14)



13 CAUTIONS & LIMITATIONS

Calibration Barrier: Full environmental mapping requires a minimum 3-day baseline to ensure patterns are statistically valid.

Sensor Constraints: Webcam gaze detection may be hindered by glasses or poor lighting, which is a useful signal, not a diagnosis.

False positives: Fast scrolling with correct answers is recognized as prior knowledge or review, not a focus failure.

Over-reliance: Learners may come to depend on the app rather than build intrinsic metacognitive skills. Mitigated by framing interventions as temporary scaffolding and offering Focus Challenges to build internal skills.

Freemium equity gap: The strongest detection (webcam gaze) is paywalled, which is a meaningful equity concern in educational deployment.

Privacy: Location, audio, and webcam require explicit user

14 THE PROMISE

*A system that sees
what the learner
cannot see about
themselves.*

For the remote learner without an instructor to notice their glazed expression, Montana offers something currently unavailable.

Every session generates evidence about whether learning actually occurred. Interventions rest on science, Bühler et al. (2024) for detection, Berman et al. (2008) for nature-walk restoration. The Focus Dashboard makes cognitive failures **visible and personal**, building the metacognitive awareness the zone-out cluster most lacks.

Turning a focus gap into a focus challenge -- passive scrolling into self-regulated learning.

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